DARK WEB THREAT INTELLIGENCE SCRAPER AND AUTOMATED THREAT CLASSIFY USING MACHINE LEARNING

SUBJECT CODE: AD22811
SUBJECT NAME: PROJECT WORK

Sri Venkateswara College of Engineering

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ABSTRACT

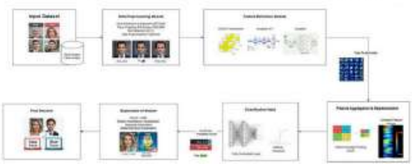
Dark Web forums and marketplaces serve as early breeding grounds for ransomware, credential leaks, and exploit sharing, leaving conventional security systems reactive and delayed. This project presents a threat intelligence system combining TOR-based dark web scraping with NLP preprocessing and ML classifiers — Naïve Bayes, Random Forest, and SVM — for accurate threat categorization. A Threat Scoring Engine prioritizes severity, while an Arduino/ESP32 IoT module delivers real-time physical alerts via LED indicators and LCD display. The system effectively shifts cybersecurity from reactive incident response to proactive, early-stage threat prevention suitable for real-world deployment.

PROBLEM STATEMENT

Cyberattacks are increasingly planned and coordinated through Dark Web forums, marketplaces, and paste sites, where threat actors exchange malware, stolen credentials, and exploit kits long before attacks are executed. Existing cybersecurity solutions operate reactively, detecting threats only after network intrusion or data compromise has occurred, resulting in significant data loss, financial damage, and delayed response. There is a critical need for a proactive, automated threat intelligence system capable of continuously monitoring dark web sources, classifying threat content using machine learning, and delivering real-time severity-based alerts to enable early intervention and preventive cybersecurity measures.

PROPOSED WORK

An end-to-end proactive cyber threat intelligence pipeline is developed that scrapes hidden .onion sources via TOR, preprocesses raw text using NLP techniques, and classifies threats using Naïve Bayes, Random Forest, and SVM classifiers.



REQUIREMENTS SPECIFICATIONS

Technology Stack: Python, Scrapy, BeautifulSoup, TOR/SOCKS5 Proxy, NLTK, SpaCy, Scikit-learn (Naïve Bayes, Random Forest, SVM), TF-IDF Vectorization, MongoDB, Matplotlib, Arduino/ESP32 IDE

Hardware: Intel i7 / AMD Ryzen 7 or higher, 16–32 GB RAM minimum, SSD ≥ 512 GB, Arduino/ESP32 Module, LEDs, Buzzer, LCD Display, GPU recommended for ML/NLP tasks

Dataset: Kaggle — CTI-Dataset / Dark Web Threat Intelligence Dataset (threat-labeled dark web forum posts and marketplace content)

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IMPLEMENTATION MODULES

Module 1 — Dark Web Scraping & Data Preprocessing Scrapes .onion sources via TOR proxy using BeautifulSoup/Scrapy, applying tokenization, stopword removal, and TF-IDF vectorization to generate clean feature vectors.

Module 2 — ML Classification & Threat Scoring Naïve Bayes, Random Forest, and SVM classifiers prioritize threats into Low / Medium / High severity using keyword weighting and source trust level.

Module 3 — IoT Real-Time Alert System Arduino/ESP32 receives threat severity via Wi-Fi and triggers Green, Yellow, or Red LED alerts with buzzer and LCD threat message.

OUTPUT SCREENSHOT

INFERENCES

- Random Forest outperforms Naïve Bayes and SVM in threat categorization by capturing complex feature interactions, making it the most reliable classifier in the ensemble pipeline.
- Threat Scoring Engine accurately prioritizes threats into Low, Medium, and High severity levels using keyword weighting and source trust scoring, significantly reducing false positive alert rates.
- Arduino/ESP32 IoT module successfully maps ML outputs to real-time physical alerts —with buzzer and LCD display – validating hardware-level feasibility.

CONCLUSION AND FUTURE WORK

Future enhancements include:

- Deep learning models such as BERT and LSTM will be incorporated to improve threat classification accuracy on obfuscated dark web text
- Integration with SIEM and SOAR platforms will automate incident response workflows based on detected threat severity.
- Model optimization for edge device deployment to enable threat detection without dependency on backend servers.
- Expanding dataset diversity with more labeled dark web threat categories to improve generalization across real-world cyberattack variations.

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